Initial and subject parameter distribution:

$$\underbrace{\begin{bmatrix} \text{T0m_bin_psx_B}_i \\ \text{T0m_con_psx_B}_i \\ \text{auto_bin_psx_W}_i \\ \text{auto_con_psx_W}_i \\ \text{auto_bin_psx_B}_i \\ \text{auto_con_psx_B}_i \\ \text{diff_bin_psx_W}_i \\ \text{diff_con_psx_W}_i \\ \text{mvaresm_psx_bin}_i \\ \text{mvaresm_psx_con}_i \\ \text{T0var_bin_psx_B}_i \\ \text{T0var_bin_psx_W}_i \\ \text{T0var_bin_psx_B_con_psx_B}_i \\ \text{T0var_con_psx_B}_i \\ \text{T0var_con_psx_W}_i \\ \text{T0MEANS_bin_psx_W}_i \\ \text{T0MEANS_con_psx_W}_i \end{bmatrix}}_{\boldsymbol{\phi}(i)} \sim \text{tform}\left\{ \text{N}\left(\begin{bmatrix} \text{T0m_bin_psx_B} \\ \text{T0m_con_psx_B} \\ \text{raw_auto_bin_psx_W} \\ \text{raw_auto_con_psx_W} \\ \text{raw_auto_bin_psx_B} \\ \text{raw_auto_con_psx_B} \\ \text{raw_diff_bin_psx_W} \\ \text{raw_diff_con_psx_W} \\ \text{raw_mvaresm_psx_bin} \\ \text{raw_mvaresm_psx_con} \\ \text{raw_T0var_bin_psx_B} \\ \text{raw_T0var_bin_psx_W} \\ \text{raw_T0var_bin_psx_B_con_psx_B} \\ \text{raw_T0var_con_psx_B} \\ \text{raw_T0var_con_psx_W} \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \text{rawPCov_1_1} & \text{rawPCov_2_1} & \text{rawPCov_3_1} & \text{rawPCov_4_1} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{rawPCov_2_1} & \text{rawPCov_2_2} & \text{rawPCov_3_2} & \text{rawPCov_4_2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{rawPCov_3_1} & \text{rawPCov_3_2} & \text{rawPCov_3_3} & \text{rawPCov_4_3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \text{rawPCov_4_1} & \text{rawPCov_4_2} & \text{rawPCov_4_3} & \text{rawPCov_4_4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \text{T0cov_2_2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \text{T0cov_4_4} \end{bmatrix} \right) \right\} + \underbrace{\begin{bmatrix} \text{raw_T0m_bin_psx_B_zpos} & \text{raw_T0m_bin_psx_B_zneg} & \text{raw_T0m_bin_psx_B_zdis} \\ \text{raw_T0m_con_psx_B_zpos} & \text{raw_T0m_con_psx_B_zneg} & \text{raw_T0m_con_psx_B_zdis} \\ \text{raw_auto_bin_psx_W_zpos} & \text{raw_auto_bin_psx_W_zneg} & \text{raw_auto_bin_psx_W_zdis} \\ \text{raw_auto_con_psx_W_zpos} & \text{raw_auto_con_psx_W_zneg} & \text{raw_auto_con_psx_W_zdis} \\ \text{raw_auto_bin_psx_B_zpos} & \text{raw_auto_bin_psx_B_zneg} & \text{raw_auto_bin_psx_B_zdis} \\ \text{raw_auto_con_psx_B_zpos} & \text{raw_auto_con_psx_B_zneg} & \text{raw_auto_con_psx_B_zdis} \\ \text{raw_diff_bin_psx_W_zpos} & \text{raw_diff_bin_psx_W_zneg} & \text{raw_diff_bin_psx_W_zdis} \\ \text{raw_diff_con_psx_W_zpos} & \text{raw_diff_con_psx_W_zneg} & \text{raw_diff_con_psx_W_zdis} \\ \text{raw_mvaresm_psx_bin_zpos} & \text{raw_mvaresm_psx_bin_zneg} & \text{raw_mvaresm_psx_bin_zdis} \\ \text{raw_mvaresm_psx_con_zpos} & \text{raw_mvaresm_psx_con_zneg} & \text{raw_mvaresm_psx_con_zdis} \\ \text{raw_T0var_bin_psx_B_zpos} & \text{raw_T0var_bin_psx_B_zneg} & \text{raw_T0var_bin_psx_B_zdis} \\ \text{raw_T0var_bin_psx_W_zpos} & \text{raw_T0var_bin_psx_W_zneg} & \text{raw_T0var_bin_psx_W_zdis} \\ \text{raw_T0var_bin_psx_B_con_psx_B_zpos} & \text{raw_T0var_bin_psx_B_con_psx_B_zneg} & \text{raw_T0var_bin_psx_B_con_psx_B_zdis} \\ \text{raw_T0var_con_psx_B_zpos} & \text{raw_T0var_con_psx_B_zneg} & \text{raw_T0var_con_psx_B_zdis} \\ \text{raw_T0var_con_psx_W_zpos} & \text{raw_T0var_con_psx_W_zneg} & \text{raw_T0var_con_psx_W_zdis} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{\boldsymbol{\beta}} \underbrace{\begin{bmatrix} \text{zpos} \\ \text{zneg} \\ \text{zdis} \end{bmatrix}}_{\boldsymbol{z}} \right\}$$

Deterministic change:

$$\underbrace{\text{d}\begin{bmatrix} \text{bin_psx_B} \\ \text{bin_psx_W} \\ \text{con_psx_B} \\ \text{con_psx_W} \end{bmatrix}(t)}_{\text{d}\boldsymbol{\eta}(t)} = \left(\underbrace{\begin{bmatrix} \text{auto_bin_psx_B} & 0 & 0 & 0 \\ 0 & \text{auto_bin_psx_W} & 0 & 0 \\ 0 & 0 & \text{auto_con_psx_B} & 0 \\ 0 & 0 & 0 & \text{auto_con_psx_W} \end{bmatrix}}_{\mathbf{A}_{\text{DRIFT}}} \underbrace{\begin{bmatrix} \text{bin_psx_B} \\ \text{bin_psx_W} \\ \text{con_psx_B} \\ \text{con_psx_W} \end{bmatrix}(t)}_{\boldsymbol{\eta}(t)} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{\mathbf{b}_{\text{CINT}}} \right) \text{d}t \quad +$$

Random change:

$$\underbrace{UcorSDtoChol\left\{ \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & \text{diff_bin_psx_W} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \text{diff_con_psx_W} \end{bmatrix} \right\}}_{\mathbf{G}_{\text{DIFFUSION}}} \underbrace{\text{d}\begin{bmatrix} W_1 \\ W_2 \\ W_3 \\ W_4 \end{bmatrix}(t)}_{\text{d}\mathbf{W}(t)}$$

Linear predictor:

$$\underbrace{\begin{bmatrix} \nu_1(t) \\ \nu_2(t) \end{bmatrix}}_{\boldsymbol{\nu}(t)} = \underbrace{\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}}_{\boldsymbol{\Lambda}_{\text{LAMBDA}}} \underbrace{\begin{bmatrix} \text{bin_psx_B} \\ \text{bin_psx_W} \\ \text{con_psx_B} \\ \text{con_psx_W} \end{bmatrix}(t)}_{\boldsymbol{\eta}(t)} + \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_{\boldsymbol{\tau}_{\text{MANIFESTMEANS}}}$$

Predicted observations:

$$\underbrace{\begin{bmatrix} \hat{Y}_1(t) \\ \hat{Y}_2(t) \end{bmatrix}}_{\widehat{\mathbf{Y}}(t)} = \begin{bmatrix} \text{logit}^{-1}\left(\nu_1(t)\right) \\ \nu_2(t) \end{bmatrix}$$

Observations:

$$\underbrace{\begin{bmatrix} \text{esm_psx_bin} \\ \text{esm_psx_con} \end{bmatrix}(t)}_{\mathbf{Y}(t)} = \widehat{\mathbf{Y}}(t) + \boldsymbol{\epsilon}(t)$$

Observation error:

$$\boldsymbol{\epsilon}(t) \sim \text{N}\left(\mathbf{0}, \underbrace{\begin{bmatrix} \hat{Y}_1(t)\left(1 - \hat{Y}_1(t)\right) & 0 \\ 0 & \left[\text{mvaresm_psx_con}\right]^2 \end{bmatrix}}_{\boldsymbol{\Theta}_{\text{effective measurement covariance}}} \right)$$

Note: a censored observation is recorded as the value above clamped to its limits, so the equation gives the uncensored mean and observations pile up at censormin and censormax rather than passing them; the error term shown is real for this kind, unlike the others, and applies to $\boldsymbol{\eta}$.

System noise distribution per time step:

$$\Delta\left[W_{j\in[1,4]}\right](t-u) \sim \text{N}(0, t-u)$$

Note: *UcorSDtoChol* converts lower tri matrix of standard deviations and unconstrained correlations to Cholesky factor,
UcorSDtoCov = transposed cross product of UcorSDtoChol, to give covariance, See Driver & Voelkle (2018) p11.
Individual specific notation (subscript i) only shown for subject parameter distribution – pop. means shown elsewhere.